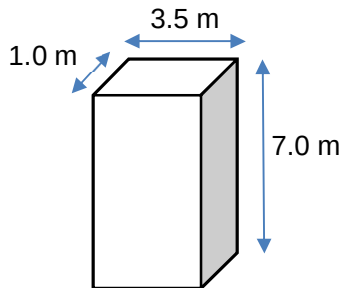


- 6 The diagram below shows a wooden block in its upright position, with length 1.0 m, width 3.5 m and height 7.0 m. The density of wood is 850 kg m^{-3} .



What is the submerged depth of the wooden block when it floats upright in water?

Density of water is 1000 kg m^{-3} .

	A	0.85 m	B	5.6 m	C	6.0 m	D	7.0 m
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Answer: C

Weight of water displaced = weight of wooden block

$$\rho_{\text{water}} g V_{\text{water displ.}} = \rho_{\text{wood}} g V_{\text{wood}}$$

$$1000 \times g \times h_{\text{water displ.}} \times A = 850 \times g \times 7.0 \times A$$

$$h_{\text{water displ.}} = 5.95 \text{ m} = 6.0 \text{ m (2 s.f.).}$$